

Practical week - 6

routes/catalog.py :

```
from fastapi import APIRouter, HTTPException
```

```
router = APIRouter()
```

```
books = [
```

```
{
```

```
    "id": 101,
```

```
    "title": "Python Programming",
```

```
    "author": "John Smith",
```

```
    "price": 450,
```

```
    "isbn": "9781234567890",
```

```
    "status": "Available"
```

```
},
```

```
{
```

```
    "id": 102,
```

```
    "title": "Database Management",
```

```
    "author": "Robert Brown",
```

```
    "price": 550,
```

```
    "isbn": "9789876543210",
```

```
    "status": "Available"
```

```
}
```

```
]
```

```
@router.get("/books") def
```

```
get_books():
```

```
    return books
```

```
@router.get("/books/{book_id}")
```

```
def get_book(book_id: int):
```

```
    for book in books:
```

```
        if book["id"] == book_id:
```

```
            return book raise
```

```
HTTPException(
```

```
        status_code=404,  
        detail="Book not found"  
    )
```

routes/admin.py :

```
from fastapi import APIRouter, HTTPException  
from pydantic import BaseModel, Field  
from routes.catalog import books  
router = APIRouter()
```

```
class Book(BaseModel):
```

```
    title: str  
    author: str  
    price: float = Field(ge=0)  
    isbn: str  
    status: str
```

```
@router.post("/admin/books") def
```

```
add_book(book: Book):
```

```
    new_id = max([b["id"] for b in books], default=100) + 1  
    new_book = {  
        "id": new_id,  
        book.model_dump()  
    }  
    books.append(new_book)  
    return {  
        "message": "Book added successfully",  
        "book": new_book  
    }
```

```
@router.put("/admin/books/{book_id}")
```

```
def update_book(book_id: int, book: Book):
```

```
    for index, existing_book in enumerate(books):  
        if existing_book["id"] == book_id:  
            updated_book = {
```

```

        "id": book_id,
        book.model_dump()
    }
    books[index] = updated_book
    return {
        "message": "Book updated successfully",
        "book": updated_book
    }

    raise HTTPException(
        status_code=404,
        detail="Book not found"
    )

@router.delete("/admin/books/{book_id}")
def delete_book(book_id: int):
    for index, book in enumerate(books):
        if book["id"] == book_id:
            deleted_book = books.pop(index)
            return {
                "message": "Book deleted successfully",
                "book": deleted_book
            }

    raise HTTPException(
        status_code=404,
        detail="Book not found"
    )

```

main.py :

```

from fastapi import FastAPI

from routes.catalog import router as catalog_router
from routes.admin import router as admin_router

app = FastAPI(
    title="BookFlow Library API"
)

app.include_router(catalog_router)
app.include_router(admin_router)

```

GET default all the data:

default

GET /books Get Books

Parameters

No parameters

Execute

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/books

Server response

Code

Details

200

Response body

```
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 499,
  "isbn": "9781234567890",
  "status": "Available"
},
{
  "id": 102,
  "title": "Database Management",
  "author": "Robert Brown",
  "price": 550,
  "isbn": "9789012345678",
  "status": "Available"
}
}
```

Response headers

```
content-length: 240
content-type: application/json
date: Tue, 15 Sep 2020 06:05:14 GMT
server: uvicorn
```

GET:

GET /books/{book_id} Get Book

Parameters

Cancel

Name	Description
book_id * required	
integer	
(path)	

101

Execute

Response:

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/books/101' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/books/101

Server response

Code

Details

200

Response body

```
{
  "id": 101,
  "title": "Python Programming",
  "author": "John Smith",
  "price": 499,
  "isbn": "9781234567890",
  "status": "Available"
}
```

Response headers

```
content-length: 117
content-type: application/json
date: Tue, 15 Sep 2020 06:22:08 GMT
server: uvicorn
```

POST:

POST /admin/books Add Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "title": "Java Programming",
  "author": "Chaitanya",
  "price": 600,
  "isbn": "9781234567891",
  "status": "Available"
}
```

Execute

Response:

Responses	
Curl	
<pre>curl -X 'POST' \ 'http://127.0.0.1:8000/admin/books' \ -H 'accept: application/json' \ -H 'Content-Type: application/json' \ -d '{ "title": "Java Programming", "author": "Chaitanya", "price": 600, "isbn": "9781234567891", "status": "Available" }'</pre>	
Request URL	
<pre>http://127.0.0.1:8000/admin/books</pre>	
Server response	
Code	Details
200	Response body <pre>{ "message": "Book added successfully", "book": { "id": 10, "title": "Java Programming", "author": "Chaitanya", "price": 600, "isbn": "9781234567891", "status": "Available" } }</pre> Response headers <pre>content-length: 160 content-type: application/json date: Tue, 15 Sep 2026 06:34:18 GMT server: uvicorn</pre>

PUT:

PUT

/admin/books/{book_id} Update Book

Parameters

Name	Description
book_id required	integer (path) 101

Request body required

Edit Value | Schema

```
{
  "title": "JavaScript Essentials",
  "author": "John Smith",
  "price": 450,
  "isbn": "9781234567890",
  "status": "Available"
}
```

Response:

Responses	
Curl	
<pre>curl -X 'PUT' \ 'http://127.0.0.1:8000/admin/books/101' \ -H 'accept: application/json' \ -H 'Content-Type: application/json' \ -d '{ "title": "JavaScript Essentials", "author": "John Smith", "price": 450, "isbn": "9781234567890", "status": "Available" }'</pre>	
Request URL	
<pre>http://127.0.0.1:8000/admin/books/101</pre>	
Server response	
Code	Details
200	Response body <pre>{ "message": "Book updated successfully", "book": { "id": 101, "title": "JavaScript Essentials", "author": "John Smith", "price": 450, "isbn": "9781234567890", "status": "Available" } }</pre> Response headers <pre>content-length: 169 content-type: application/json date: Tue, 15 Sep 2026 06:38:50 GMT server: uvicorn</pre>

DELETE:

DELETE

/admin/books/{book_id} Delete Book

Parameters

Cancel

Name	Description
book_id required	
Integer (path)	102

Execute

Response:

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8080/admin/books/102' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8080/admin/books/102
```

Server response

Code

Details

200

Response body

```
{
  "message": "Book deleted successfully",
  "book": {
    "id": 102,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 550,
    "isbn": "9780130643210",
    "status": "Available"
  }
}
```

Download

Response headers

```
content-length: 167
content-type: application/json
date: Tue, 15 Sep 2026 06:30:35 GMT
server: uvicorn
```